Worksheet — Friction on an Incline Simulation

Use with *Simulation - Friction on an incline.html* · goes with the Forces investigation (friction on a ramp)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/03-friction-incline** in a browser (offline: *Simulation - Friction on an incline.html*). ▲ **Tilt** raises the ramp until the block slips; ▶ **Release lets the block go at the angle in the box.** The **Tasks** tab has the same three parts as this sheet; write your **explanations** here.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use g = 9.81 N kg^{-1}. Sliding uses Newton's 2nd law (Blocks 3–4).

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. The ramp tilts and the block stays at rest. What happens to the friction force?			
2. What happens to the normal force as the ramp tilts?			
3. You double the block's mass. What happens to the slip angle?			
4. Which surfaces give the smallest slip angle: rubber, wood/bench, ice or sandpaper?			
5. After it slips (ramp angle fixed), does the block slide at constant speed, speed up or stop?			

6. Draw the free-body diagram of the block at rest on a 20° ramp. Show the weight components $mg \sin\theta$ and $mg \cos\theta$ as dashed arrows.

Free-body diagram

7. Show that the block slips when $\tan\theta = \mu_s$. Then calculate the slip angle for wood on the bench ($\mu_s = 0.41$).

Working

θ_{slip} = _____ ° Sim check: _____ °

Part 2 — During the lab: Forces investigation

Investigation block (0.150 kg) on the wooden board. Fill in the sim values and your group's values.

Quantity	Simulation	Our measurement / calculation
slip angle θ_{slip} / °		
$\mu_s = \tan\theta_{slip}$		
largest static friction at 20°, $\mu_s mg \cos 20°$ / N		
friction acting at 10°, $mg \sin 10°$ / N		
acceleration at 60°, a / m s^{-2}		7.8 (given)
$\mu_d = (g \sin\theta - a)/(g \cos\theta)$		

8. Give **one** reason why your measured slip angle differs from the simulation's.

9. At 10° the friction is much smaller than $\mu_s F_N$. Explain why.

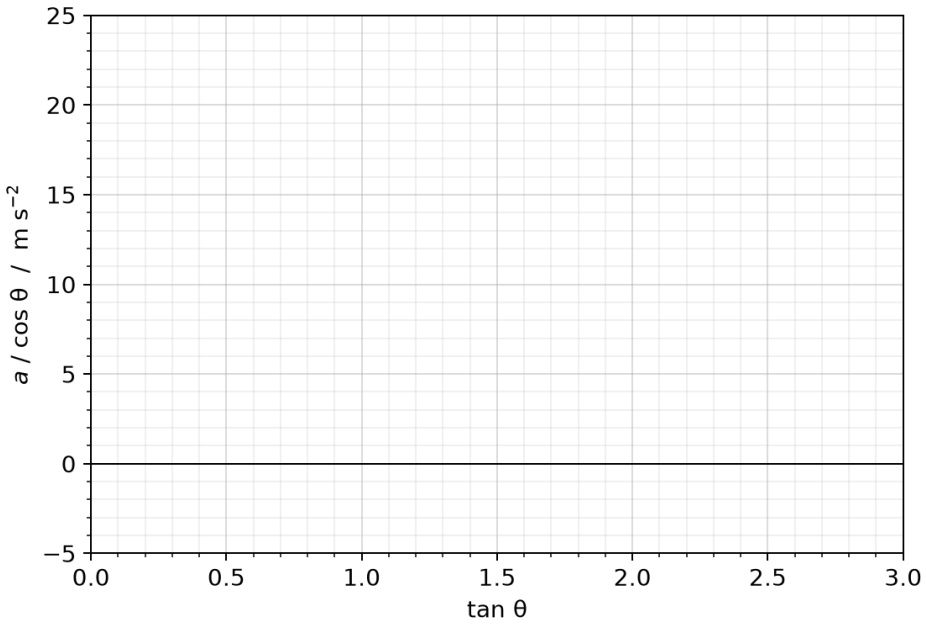
Part 3 — Homework: find μ_d from a graph

Wood on bench, m = 0.500 kg. Release the block at each angle and record a.

θ / °	30	40	50	60	70
tan θ					
a / m s^{-2}					
cos θ					
a / cos θ / m s^{-2}					

10. Show that $a = g(\sin\theta - \mu_d \cos\theta)$ can be written as $a/\cos\theta = g\tan\theta - \mu_d g$. What should the gradient and the intercept of a graph of $a/\cos\theta$ against $\tan\theta$ be?

11. Plot $a/\cos\theta$ against $\tan\theta$ and draw the best-fit line.



12. Gradient = _____ m s^{-2} Intercept = _____ m s^{-2} ⇒ μ_d = _____

13. Calculate the acceleration of the block sliding down the bench at 35° (μ_d = 0.30) and the time it takes to slide 0.80 m from rest.

Working

14. The 0.500 kg block is released on the bench at 15° (μ_s = 0.41). Draw its free-body diagram. Does it move? Calculate the friction force acting and compare it with $\mu_s F_N$.

Free-body diagram

Working

15. A car is parked on a steep hill. On a dry road it stays put; on ice it slides down. Explain why, using tan θ and μ_s. Does a heavier car slide more easily?